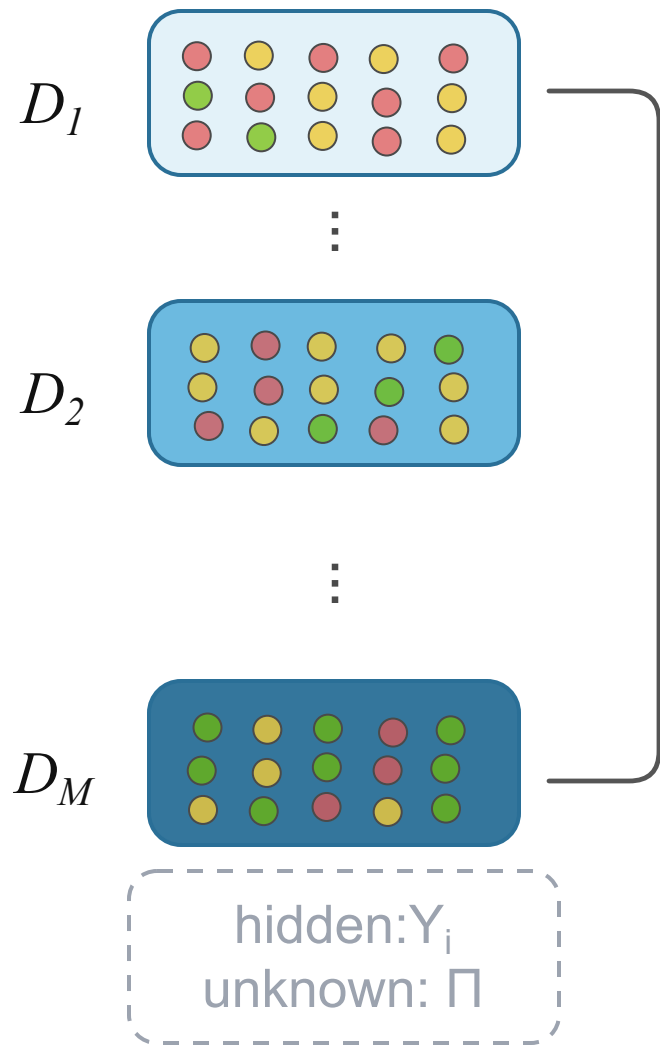


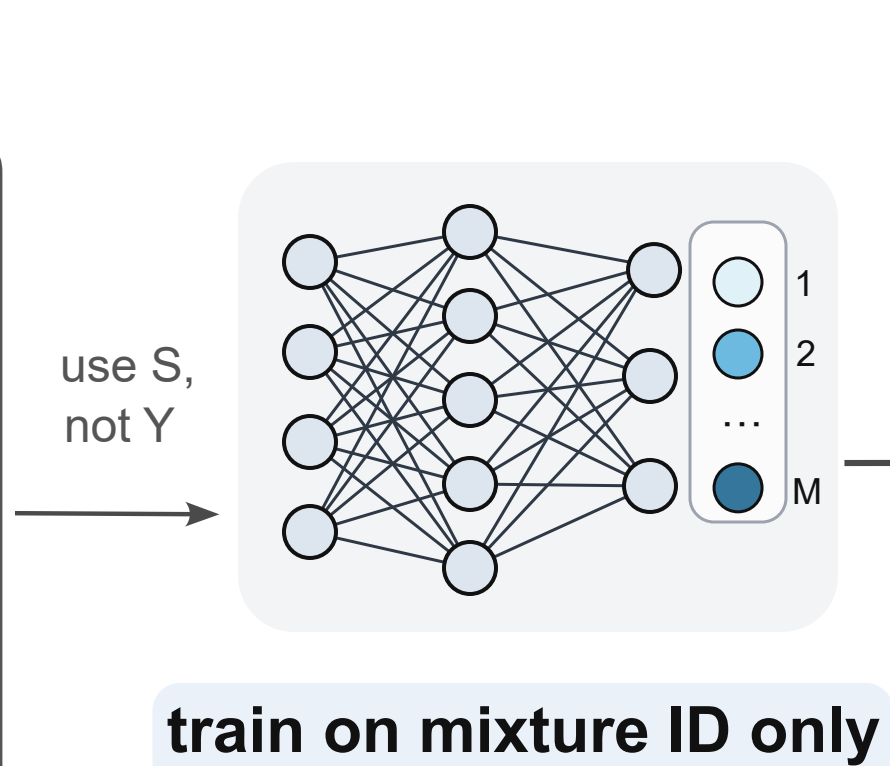
Unlabeled mixtures

observed: x_i , $S_i = m$



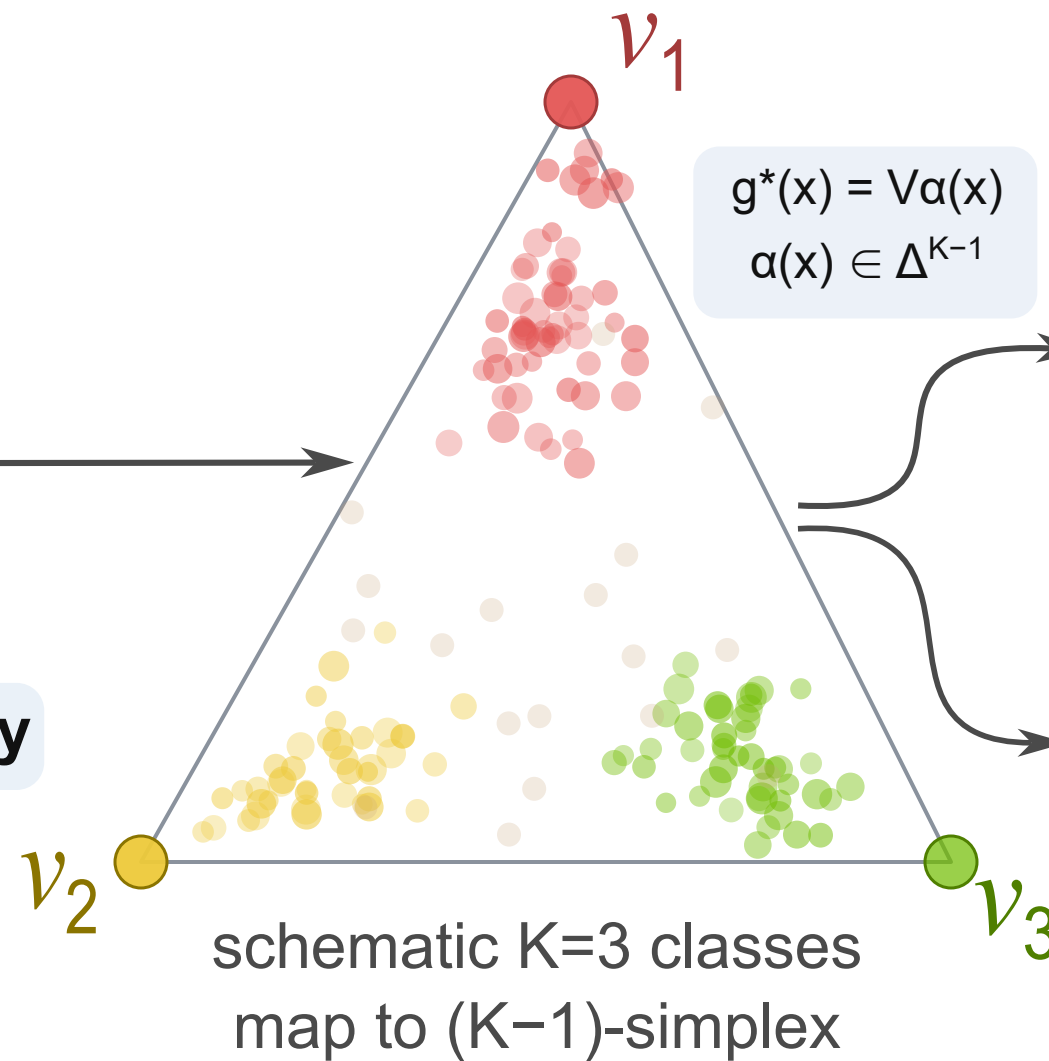
M-way mixture classifier

$$g_\theta(x) \approx P(S \mid X=x)$$



Mixture-posterior simplex

$$\{g_\theta(x_i)\} \subset \Delta^{M-1}$$



Prior-free latent recovery

R1: post-hoc simplex fit

- estimate $\hat{V}$
- barycentric decode $\alpha_\theta(x)$

R2: bottleneck head

$$g_\theta(x) = \hat{\Pi} \alpha_\theta(x)$$

latent predictor

$$\hat{\alpha}(x)$$

